

RESAVO

Every Resource. A Better Next Use.

Full App, Platform, Prototype, Architecture & Build Specification

Created by merging the strongest mechanisms from SIH 26234 and REGAIN OS, then extending them beyond the original limits into a people-first public-benefit platform.

Working name: RESAVO (name is a design choice; trademark/domain clearance is still required before public launch.)

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1. Executive definition

RESAVO is a people-first digital network that helps people and organizations offer useful resources, helps other people and organizations find what they genuinely need, and uses intelligence to decide whether a transfer is actually practical. Its success is measured by verified resources preserved, reused, transferred or recovered - not by transaction count alone.

RESAVO - Offer, Need and Impact Network

The system does not force a transaction; it finds the best feasible next use.



Figure 1 - RESAVO connects offers, needs and real-world feasibility instead of acting as a blind marketplace.

The project starts from two foundations: the SIH 26234 food-waste problem and the REGAIN closed-loop mechanism. SIH 26234 contributes food-demand prediction, surplus identification, quality monitoring, redistribution, logistics, processing efficiency and sustainability intelligence. REGAIN contributes the deeper loop: capture -> predict loss -> intervene -> verify outcome -> measure recovery -> learn.

The redesigned product is intentionally not an SIH submission. SIH is only the source problem that gives the project a concrete first domain. The product itself can exceed that scope and is designed as a public-benefit application for people, families, businesses, NGOs, schools, institutions and communities.

2. What was merged and what was changed

Source foundation	Strong ideas retained	How RESAVO transforms them
SIH 26234	Demand forecasting; surplus detection; food quality; redistribution; logistics; processing efficiency; sustainability	These become the Food Intelligence and network coordination layers.
REGAIN OS	Observe-predict-intervene-measure-learn; risk scoring; intervention hierarchy; outcome verification; recovery value; resource passport	These become the universal intelligence and trust loop for every supported item.
New product layer	Seller + Buyer + Admin roles; purpose-aware matching; alternative-supply checks; transfer feasibility; public-benefit impact	This creates a real operational network rather than a tracker or dashboard.

Key design change: RESAVO does not assume that every offer should be matched to the nearest buyer. It first asks whether the requested transfer is useful and practical compared with the buyer's normal alternatives.

3. Product vision and non-negotiable principles

People first: The product is built to help users save, access, reuse, share or transfer useful resources. Monetization is not the project objective.

Public-benefit operation: No paid ranking, advertising-driven recommendations, transaction commission or sale of user resource history. The system exists to maximize practical public benefit.

Offer is not transaction: An offer is simply availability. It becomes a transaction only after a feasible need and workable fulfillment path are found.

Best next use, not maximum trading: The system should sometimes say: do not transfer, use it yourself, use a local supplier, donate, combine offers, or wait.

Verification before impact: A listing, match or recommendation is not counted as impact until the relevant outcome is confirmed.

Simple for humans, complex underneath: Users should see one clear action. Complexity stays inside the decision engine.

Privacy by default: Only the minimum data needed for matching, safety and analytics is collected. Household histories are not public.

Food first, electricity later: Electricity monitoring is explicitly deferred. Future integrations are planned, but Phase 1 does not depend on them.

Resource-specific evidence: Each item type uses its own evidence method: food uses quantity/time/quality context; clothing uses image + user confirmation; books use category/condition; etc.

4. Portals and roles

RESAVO uses exactly three primary product portals. A transfer partner does not require a fourth portal in the initial build; pickup/drop tasks can be handled inside the Seller or Buyer mobile app using a temporary transfer-task role.

Portal	Who uses it	Primary device	Purpose
Seller / Provider Portal	Families, shops, restaurants, hotels, schools, institutions, small businesses	Mobile app	Create offers, set terms, see matches, confirm handover, track impact.
Buyer / Receiver Portal	Families, businesses, NGOs, schools, hostels, community groups, approved recipients	Mobile app + responsive web	Create needs, compare sources, accept offers, coordinate receipt and confirm outcome.
Admin / Impact Console	Platform administrators and trusted operations staff	Web	Verify organizations, moderate users, monitor network growth, measure verified impact, resolve disputes, manage rules.

Role switching: one account may have both Seller and Buyer capabilities. For example, a bakery can buy milk and also offer surplus bread. The UI therefore uses a role switch rather than separate accounts.

4.1 Permission model

Capability	Seller / Provider	Buyer / Receiver	Admin
Create offers	Yes	No	Can moderate only
Create needs	Optional	Yes	Can view aggregate
Accept/reject match	Yes	Yes	Can intervene for disputes
Confirm pickup	Yes / transfer-task role	Yes / transfer-task role	Can audit
Confirm receipt	No	Yes	Can audit
View personal impact	Own data only	Own data only	Aggregate + verified system impact
Manage organizations	No	No	Verify / suspend / configure
Manage taxonomy and rules	No	No	Yes with audit log
Resolve disputes	No	Can submit evidence	Yes

5. Phase 1 item catalog

Phase 1 deliberately focuses on resources that can be offered and needed using reliable phone-based evidence. Electricity, water-meter automation, medicines and hazardous materials are not part of the first build.

5.1 Evidence and eligibility rules

Item family	Minimum evidence	Eligibility check	Typical failure reason
Vegetables / fruits	Photo + quantity + availability window	Condition visually plausible; seller confirmation; time window	Too close to deterioration / insufficient quantity
Milk / dairy	Quantity + storage/handling confirmation + time window	Handling requirements satisfied; receiver can use in time	Unsafe/unclear handling or late transfer
Bakery / packaged food	Photo + package condition + date metadata when applicable	Packaging and date information available	Damaged package / unknown condition
Clothing	Photo + condition + size	Visible condition acceptable; seller confirms hidden damage	Severe damage / missing information
Books / school items	Photo + category + condition	Readable/usable condition	Too damaged / incomplete
Household reusable	Photo + dimensions + condition	Safe and usable; no hazardous material	Broken / unsafe / prohibited item

Category	Examples	Primary evidence	Typical demand
Fresh food	Vegetables, fruits, leafy greens	Freshness/time window; quantity; condition	Families, restaurants, hostels, community kitchens
Dairy	Milk, curd, paneer and other locally permitted dairy items	Quantity; storage/handling confirmation; time window	Families, bakeries, restaurants, institutions
Bakery / packaged food	Bread, buns, biscuits, sealed packaged food	Expiry/use-by where applicable; package condition	Families, shops, institutions, NGOs where suitable
Staples / ingredients	Rice, flour, pulses, grains, cooking ingredients	Quantity; package condition; shelf-life context	Families, restaurants, hostels, community kitchens
Prepared surplus food	Eligible cooked/ready food from institutions	Preparation time; handling; recipient eligibility	Approved organizations / community channels
Clothing	Shirts, trousers, dresses, sarees, uniforms, jackets	Photo + condition + size	Families, NGOs, shelters, schools
Warmth / bedding	Blankets, sweaters, bedsheets	Photo + condition + size	Families, shelters, community groups
Footwear	Shoes, sandals, school shoes	Photo + size + condition	Families, NGOs, schools
Education	Books, notebooks, stationery, school bags	Photo + category + condition	Students, schools, libraries, NGOs
Household reusable	Utensils, containers, storage items, non-electrical furniture	Photo + condition + dimensions	Families, hostels, community groups
Business / institutional surplus	Excess non-hazardous inventory, pantry supplies, reusable event/office items	Quantity + category + condition	Other businesses, institutions, NGOs

6. Seller / Provider experience

Seller experience is designed around one simple sentence: "I have something useful that I may not need." The seller should be able to create an offer in under a minute for ordinary items.

6.1 Seller home

- At-risk offers: items likely to become unusable or lose value soon.
- Active offers: quantity, price/free/donation mode, status and expiry of the offer.
- Matches: accepted or recommended receivers.
- Transfer tasks: pickup/drop responsibility and verification.
- Impact: quantity preserved and verified value preserved.

6.2 Create offer

Field	Required?	Notes
Category / item	Yes	Search-first taxonomy; camera can suggest a category.
Quantity + unit	Yes	kg, litre, piece, pair, box, packet etc.
Condition	Yes for non-food	New / good / usable / needs repair / not usable.
Offer mode	Yes	Sell, low-price, free, donate, exchange.
Expected price	Conditional	Required only when seller chooses paid mode.
Available until	Yes for perishable	Time window for food and time-sensitive items.
Photos	Recommended	Required for clothing/household in many flows.
Pickup preference	Yes	Seller pickup, buyer pickup, either.
Location	Approximate by default	Exact pickup location revealed only after acceptance.

6.3 Seller control

- Approve or reject a proposed receiver.
- Set minimum acceptable price where selling is selected.
- Choose "I prefer a nearby receiver" or "I prefer best-value option".
- Choose whether community/NGO receivers should be prioritized.
- Cancel before handover.

7. Buyer / Receiver experience

The Buyer portal is not a catalog-only marketplace. A buyer creates a real need, with quantity and time requirements, and RESAVO compares multiple ways of fulfilling that need.

7.1 Buyer / Receiver types

- Individual / family
- Business / bakery / restaurant / shop
- NGO / charity / community group
- School / hostel / institutional kitchen
- Approved recipient organization

7.2 Create need

Field	Example
Item	Milk
Quantity	3 kg
Need by	5:00 PM
Purpose	Bakery production
Preferred mode	Buy / low-price / free
Location	Bakery
Fulfillment preference	Pickup / delivery / either

7.3 Compare sources

The buyer sees a ranked list of practical options, not simply sellers sorted by distance. The explanation under each option is mandatory so that the recommendation is understandable.

8. Admin / Impact Console

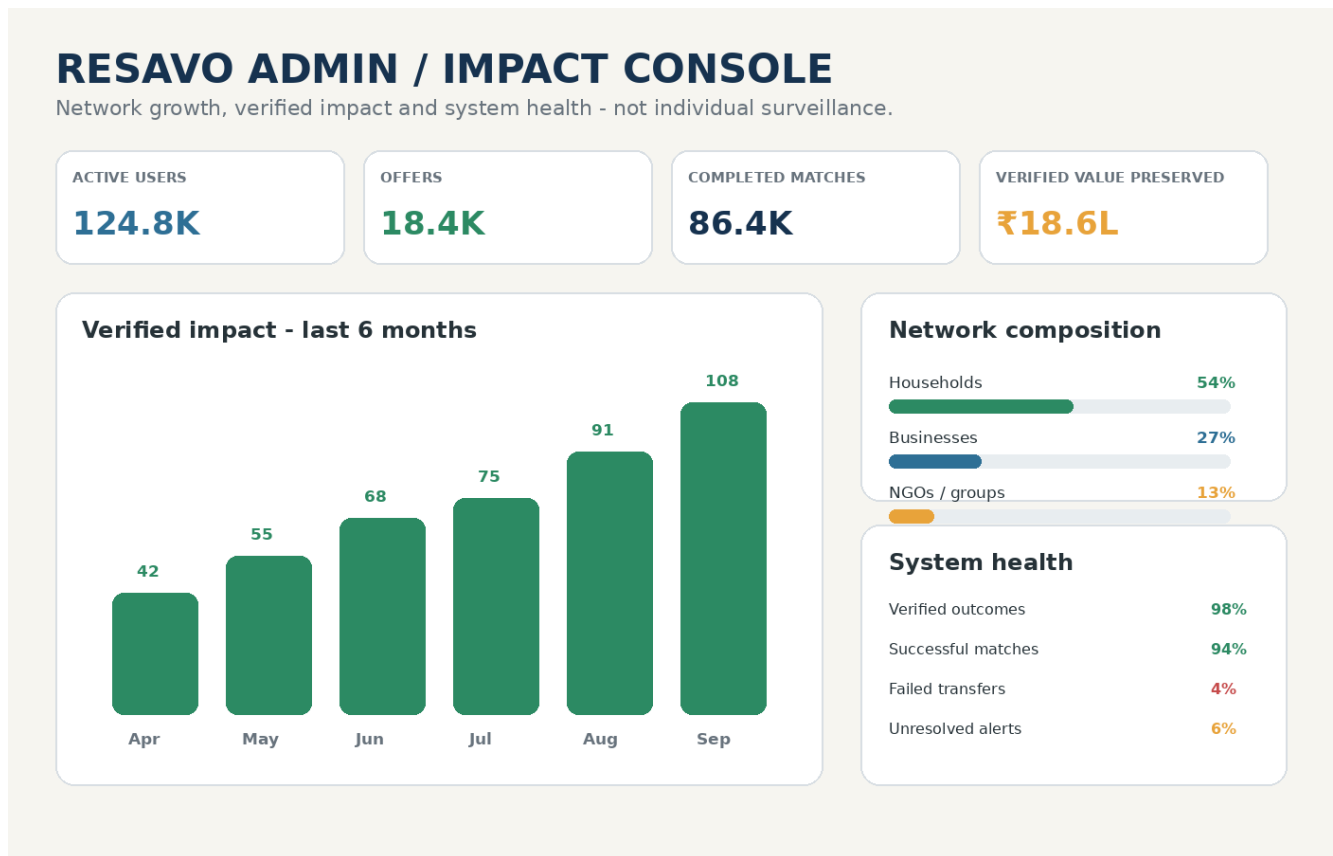


Figure 2 - Prototype visual direction for the Admin / Impact Console.

8.1 Admin modules

Module	What admin sees
Network Growth	Active users, new users, active sellers, active buyers, organizations, communities, geographic coverage.
Impact	Resources preserved, completed transfers, verified value preserved, avoided loss, community needs fulfilled.
Operations	Open matches, transfer delays, failed handovers, unresolved disputes, stale offers.
Trust & Verification	Verified organizations, flagged listings, duplicate accounts, abuse signals, suspicious patterns.
Taxonomy & Rules	Item categories, handling rules, eligibility rules, offer/need modes, intervention templates.
Analytics	Trend analysis by category, region, user type and fulfillment mode without exposing private household histories.
Reports	Daily, weekly and monthly impact exports; incident reports; system-quality reports.

8.2 Admin KPIs

KPI	Definition
Verified Value Preserved	Value supported by a verified outcome, not merely by a listing or recommendation.
Successful Match Rate	Accepted matches that reach a verified handover.
Transfer Feasibility Rate	Proposed transfers that were actually practical under the stated rules.
Offer-to-Outcome Time	Time from offer creation to confirmed outcome.
Unnecessary Transfer Avoidance	Cases where the system correctly chose a local/alternative option instead of a wasteful transfer.
Community Need Fulfilment	Verified needs successfully satisfied through the network.

9. Core matching and feasibility engine

RESAVO - End-to-End Transaction + Impact Flow



Decision principles

- A match is accepted only if the real-world transfer is practical.
- The system may recommend "do not transfer" when a local alternative is better.
- Seller intention (sell / low-price / free / donate) is respected.
- Buyer purpose (personal / business / NGO / institution) is part of matching.
- Impact is counted only after outcome verification.

Figure 3 - The end-to-end decision pipeline.

The matching engine is the key technical differentiation. A naive marketplace asks "Who is nearby?" RESAVO asks "Which feasible pathway creates the most useful outcome, considering alternatives?"

9.1 Match decision factors

Factor	Question
Need fit	Does the receiver actually need this exact item, quantity and condition?
Alternative supply	Does the receiver have a closer or easier source?
Time window	Can the item be received before it becomes unsuitable?
Distance / route burden	How much movement is required?
Fulfillment	Who can realistically pick up or deliver?
Offer terms	Sell, low-price, free, donation or exchange?
Purpose	Personal use, business input, NGO distribution, institutional use?
Reliability	Have the parties completed similar handovers successfully?
Safety / eligibility	Are there handling or recipient rules?

9.2 Recommendation outcomes

- MATCH: a practical pairing exists.
- ALTERNATIVE: a better local or existing supply is available.
- POOL: combine several nearby offers into one fulfillment run.
- WAIT: timing is not yet right and the item remains safe/useful.
- NO TRANSFER: movement would create more burden than benefit or is not safe/feasible.

10. Transfer and fulfillment logic

RESAVO does not own a fleet. In the first build, fulfillment is handled using the least disruptive available method: buyer pickup, seller drop-off, a verified transfer route, or a combined pickup run. A transfer should not be initiated only because a match exists.

Priority	Fulfillment method	When used
1	Buyer pickup	Buyer already has staff/vehicle nearby.
2	Seller drop-off	Seller is already travelling near the buyer.
3	Existing route piggyback	A verified existing route can carry the item without creating a separate trip.
4	Pooled local pickup	Multiple nearby offers can be collected together.
5	Verified transfer partner	Only when the item, timing and route requirements are satisfied.
6	No transfer	If movement is inefficient, unsafe or too late.

10.1 Verification

- Pickup confirmation: sender or pickup actor confirms quantity and condition.
- Receipt confirmation: receiver confirms quantity received and accepted condition.
- Evidence: optional photo, timestamp and location proof.
- Outcome state: complete / partial / failed / cancelled, with reason.
- Impact state: only verified outcomes enter the verified impact ledger.

11. Food-specific intelligence from SIH 26234

Food is the deepest Phase 1 domain because the SIH 26234 brief already provides a strong operational context: demand forecasting, surplus identification, quality monitoring, redistribution, logistics, processing efficiency, resource use and sustainability analytics.

11.1 Food lifecycle

Forecast -> Procure -> Store -> Prepare -> Produce -> Serve -> Surplus -> Redistribute / Reuse / Recover -> Verify -> Learn

11.2 Food intelligence features

Feature	Output
Demand forecasting	Expected quantity by time window, with confidence range.
Surplus prediction	Expected surplus quantity and time window.
Quality / usability context	Condition state based on source data and human confirmation; no unsafe AI certainty.
Production planning	Suggested production adjustment to reduce overproduction.
Redistribution matching	Eligible organizations and practical fulfillment paths.
Route planning	Pickup/transfer sequence that respects time and handling constraints.
Processing efficiency	Input-output variance, unusual losses, downtime signals where data is available.
Outcome verification	Consumed / transferred / reused / recovered / lost quantities.
Food impact	Quantity preserved and verified value preserved.

12. AI / ML design

AI is an implementation layer, not the product itself. RESAVO should begin with explainable rules and small models, then move toward stronger machine learning as verified outcome data grows.

12.1 AI modules

Module	Initial approach	Later approach
Item classification	Image model + taxonomy rules	Fine-tuned classification model per category.
Demand forecast	Moving average / gradient boosting baseline	Time-series model with event/season/context features.
Loss-risk score	Deterministic scoring + calibration	Supervised probability model.
Match ranking	Weighted scoring	Learning-to-rank after sufficient outcomes.
Route / transfer choice	Rule + shortest-path constraints	Constraint optimization / operations research.
Anomaly detection	Statistical thresholds	Isolation Forest / robust anomaly models.
Intervention learning	Outcome success rates	Contextual ranking / policy optimization with safety constraints.

12.2 Confidence policy

- Every AI result carries confidence and evidence source.
- The system shows a range where exact measurement is unavailable.
- Low-confidence results trigger user confirmation or an alternate workflow.
- High-risk safety decisions require verified rules or human confirmation.

13. Data model and database

PostgreSQL is the system of record. PostGIS can be used for geographic queries and distance checks. Object storage holds photos/evidence. Redis handles short-lived state, queues and scheduled work.

Table	Core fields
users	id, role_flags, phone/email, locale, consent_status, status
profiles	user_id, type, service_area, preferences, verification_state
offers	id, provider_id, item_id, quantity, unit, terms, available_until, location, status
needs	id, requester_id, item_id, quantity, needed_by, purpose, fulfillment_preference, location, status
items	id, category, subcategory, canonical_name, handling_class
item_evidence	item_id, photo_ref, source_type, condition, confidence, notes
matches	id, offer_id, need_id, score, decision, explanation, created_at
transfers	id, match_id, mode, pickup, dropoff, planned_time, status
verifications	id, transfer_id, actor_id, event_type, quantity, timestamp, evidence_ref
impact_events	id, source_event, quantity_preserved, value_preserved, confidence, verified
organizations	id, type, name, verification_state, service_area, capacity
disputes	id, related_object, reporter, reason, resolution
audit_logs	actor, action, object, timestamp, metadata
model_runs	model_version, dataset_version, metrics, timestamp

14. System architecture

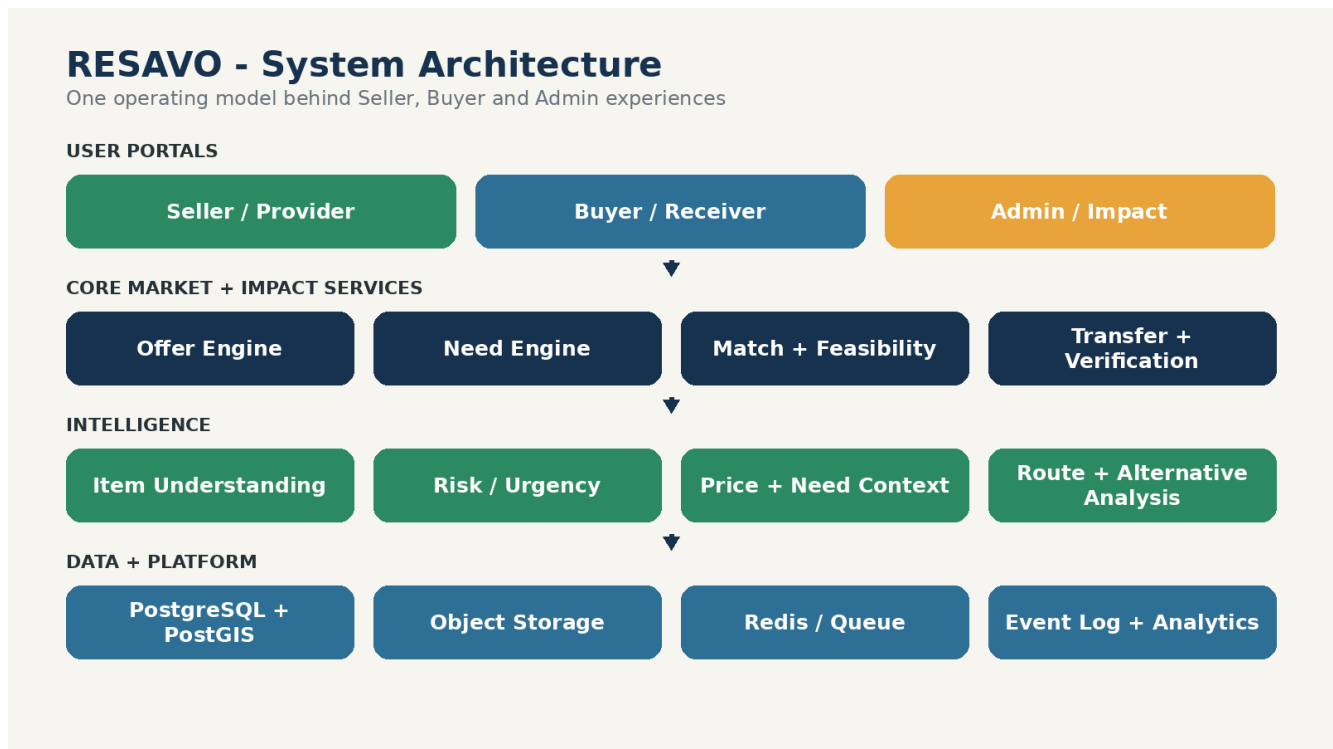


Figure 4 - High-level technical architecture.

14.1 Recommended technology

Layer	Technology	Why
Mobile app	Flutter + Dart	Single codebase for Android/iOS, camera, notifications and offline-first flows.
Web admin	Next.js + React + TypeScript	Fast responsive dashboards and role-based web experiences.
Backend API	FastAPI + Python	Strong fit for validation, business logic and ML integration.
Database	PostgreSQL + PostGIS	Transactions, integrity, analytics and geographic queries.
Cache / jobs	Redis	Queues, reminders, short-lived state and background tasks.
Storage	S3-compatible object storage	Photos, receipts and transfer evidence.
Maps / routing	OpenStreetMap data + routing engine	Avoid lock-in; suitable for distance and route feasibility.
Auth	JWT/OAuth + OTP option	Secure role-based access and convenient sign-in.
Notifications	FCM / APNs + SMS fallback	Push notifications plus fallback for low-connectivity users.
Deployment	Docker + cloud VM/container platform	Repeatable deployment and scaling.
Monitoring	OpenTelemetry + Grafana / managed logs	Observability, performance and failures.
CI/CD	GitHub Actions	Build, test, lint, security checks and deployment automation.

15. UI/UX and visual system

The product should look like a trustworthy public utility - not a flashy AI demo and not a generic ecommerce marketplace. Use clean cards, strong typography, subtle borders and high-contrast action colors.

Token	Value	Use
Primary Navy	#16324F	Headers, navigation, trust elements
Impact Green	#2C8A63	Save/verified/success actions
Route Blue	#2E6F95	Buyer actions, information, links
Caution Amber	#E8A33A	At-risk, pending, time-sensitive states
Alert Red	#C64B4B	Failed, unsafe, dispute states
Background	#F6F5F0	Warm neutral canvas
Card	#FFFFFF	Primary surfaces
Border	#D8DEE3	Low-noise separators

15.1 Design rules

- No gradients as a default visual language.
- No dark neon "AI dashboard" look.
- Every screen has one primary action.
- Use item photos and simple icons for quick recognition.
- Use human language: "3 kg milk available" instead of "resource entity created".
- Use status labels that can be understood in one glance.
- Support English + Hindi first; structure the app for more languages later.

16. Prototype screens

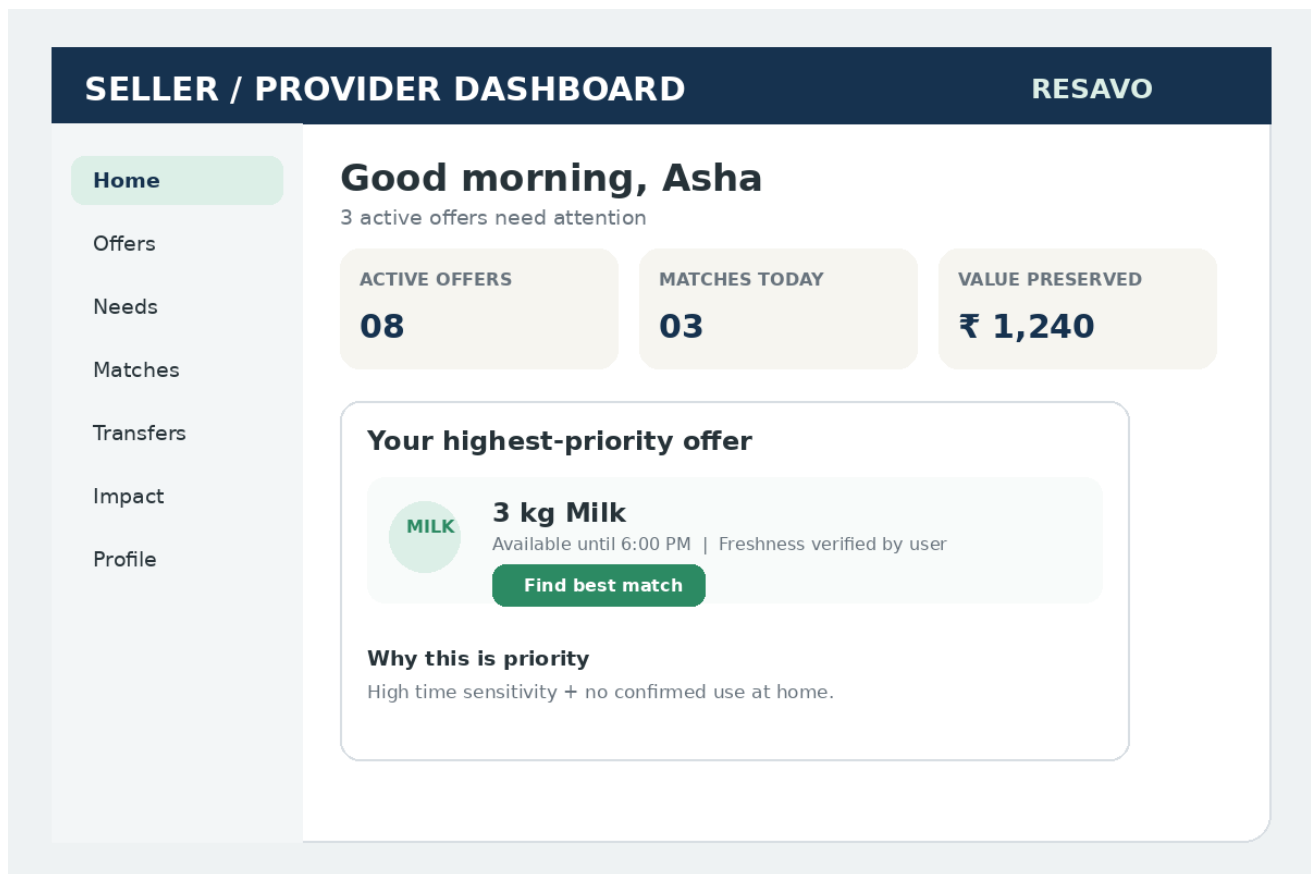


Figure 5 - Seller / Provider dashboard direction.

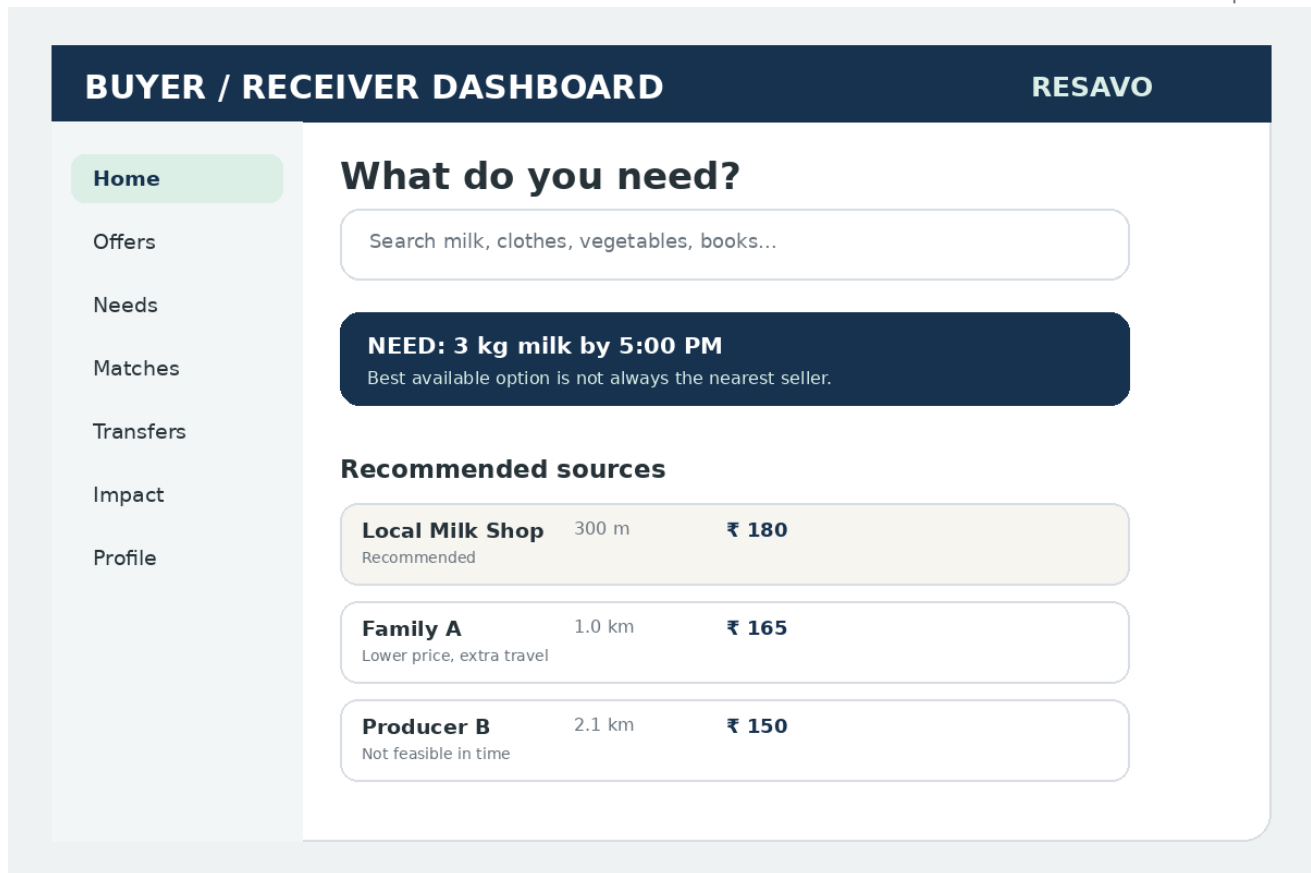


Figure 6 - Buyer / Receiver dashboard direction.

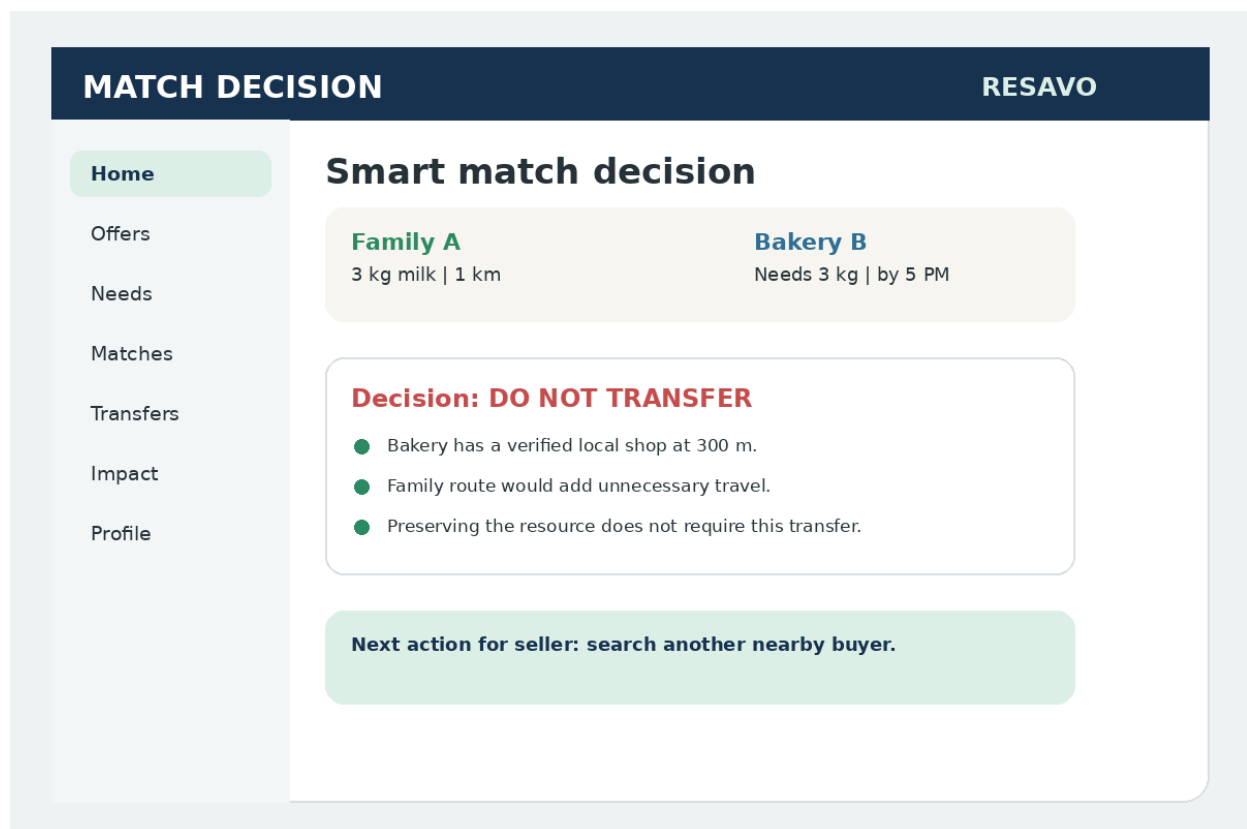


Figure 7 - Smart match decision direction; note the system can recommend not transferring.

16.1 Screen-by-screen specification

Screen	Portal	Must contain	Primary action	Important states
Auth / role selection	All	Phone/email, OTP or secure login, language, role switch	Continue	New user; verified user; suspended user
Seller Home	Seller	Active offers, at-risk offers, recommended next action, verified value preserved	Add offer / act on priority	No active offers; urgent item; pending match
Create Offer	Seller	Item, category, quantity, condition, offer mode, price if relevant, availability window, pickup preference, location	Publish offer	Draft; validation error; offline queued
Offer Detail	Seller	Evidence, matched needs, decision reasons, transfer plan, status timeline	Accept / reject / edit	Unmatched; matched; transfer pending; completed
Buyer Home	Buyer	Active needs, recommendations, nearby options, saved needs	Add need / review source	No results; alternative supply; urgent need
Create Need	Buyer	Item, quantity, deadline, purpose, acceptable condition, pickup/delivery preference	Find options	Draft; partial match; no feasible source
Source Comparison	Buyer	Seller/shop alternatives, distance, price/free status, fulfillment, route burden, reason	Choose source	Recommended; alternative; not feasible
Match Decision	Seller + Buyer	Side-by-side offer/need, feasibility explanation, alternative check	Accept / reject	MATCH; ALTERNATIVE; POOL; WAIT; NO TRANSFER
Transfer Tracking	Seller + Buyer	Pickup, ETA/window, handler, proof, status timeline	Confirm pickup / receipt	Assigned; in transit; partial; disputed; completed
Impact	Seller + Buyer	Quantity preserved, value preserved, verified outcomes, history	View evidence	Estimated vs verified clearly separated
Admin Overview	Admin	Growth, impact, operations health, alerts	Open module	Healthy; degraded; incident
Admin Verification / Disputes	Admin	Organizations, evidence, flags, audit trail	Verify / resolve	Pending; approved; suspended; resolved

Prototype rule: every interactive screen must have loading, empty, success, error and offline states. The primary action should be obvious without scrolling when practical.

17. End-to-end examples

17.1 Milk: family -> bakery

Family offers 3 kg milk for sale. Bakery needs 3 kg by 5 PM. A local milk shop 300 m away can already satisfy the bakery need, while the family is 1 km away. RESAVO compares the alternatives and recommends the local shop. The family offer is not forced into a transaction. The seller is then shown other nearby demand opportunities.

17.2 Milk: family -> bakery, when a transfer is sensible

If the local shop is unavailable or much farther away, and the bakery has a verified nearby route passing the family, RESAVO can recommend that existing route. The transfer is accepted only when time, handling and route constraints are satisfied.

17.3 Clothes: family -> NGO at low price

A family offers 20 good-condition clothes at a low price rather than free. An NGO needs bulk clothing for community distribution. RESAVO recognizes the NGO as a purpose-specific receiver, checks quantity and condition, evaluates fulfillment, and can suggest the NGO as a strong match even if another buyer offers a slightly higher price - depending on seller preferences and feasibility.

17.4 Vegetables: household -> restaurant

A household has 5 kg vegetables likely to remain unused. A restaurant needs 8 kg but already has a nearby supplier at 250 m. RESAVO may recommend no transfer. If the supplier is unavailable and the household is on the route of a normal collection trip, the offer can become useful.

17.5 Surplus cooked food

An institutional kitchen predicts 100 kg surplus. The system checks whether production can still be reduced; if not, it evaluates eligible redistribution pathways, receiving capacity and transport time. The outcome is confirmed after handover. Only the verified quantity is added to impact metrics.

17.6 Books

A family has school books no longer needed. A nearby school or NGO may have a need. The system can support free transfer, low-price sale or donation, with photo/condition confirmation and receiver verification.

18. Safety, trust and anti-abuse

18.1 Food safety

- The app should never claim that a food item is safe to eat purely from an image.
- Use handling/time/temperature inputs where available and explicit organization rules.
- Eligible redistribution pathways should be configurable by authorized administrators.

18.2 Fraud / abuse prevention

- Rate limits and duplicate listing detection.
- Photo similarity and repeated content checks for suspicious listings.
- Organization verification before high-volume or institutional receiving.
- Transaction dispute workflow with evidence and audit trail.
- Location fuzzing on public listings; exact location revealed only after acceptance.

18.3 Privacy

- Minimal personal data collection.
- Role-based access control.
- Encrypted transport and storage.
- Household histories excluded from community analytics.
- Account deletion and data export.

19. Implementation plan

Phase	Build focus	Deliverable
Phase 0	Product definition	Final taxonomy, UX flows, data model, safety rules.
Phase 1	Core app	Auth, Seller, Buyer, Admin shells; offers and needs CRUD.
Phase 2	Match engine	Distance, alternative-source checks, scoring and explainable recommendations.
Phase 3	Transfer	Pickup/drop tasks, verification, status lifecycle, dispute handling.
Phase 4	Food intelligence	Demand forecast baseline, surplus risk, food-specific workflows.
Phase 5	Impact	Verified value preserved ledger, admin analytics, network growth.
Phase 6	ML upgrade	Train models from verified outcomes; calibration and intervention ranking.
Phase 7	Scale	Offline-first sync, regional configuration, observability, multilingual support.

19.1 Suggested first build scope

The first prototype should fully support three item families: vegetables, milk/dairy, and clothing. This gives one highly perishable domain, one everyday food domain and one durable non-food domain. It should also support NGO receivers so the clothing use case is demonstrated properly.

20. API and application contract

The following is a concrete implementation contract for an AI coding assistant or development team.

Endpoint	Purpose
POST /auth/send-otp	Start login
POST /auth/verify-otp	Create/login session
POST /offers	Create seller offer
GET /offers/me	Seller offers
POST /needs	Create buyer need
GET /needs/me	Buyer needs
GET /matches/suggestions	Get explainable match candidates
POST /matches/{id}/accept	Accept match
POST /matches/{id}/reject	Reject match
POST /transfers	Create transfer task
POST /transfers/{id}/pickup	Confirm pickup
POST /transfers/{id}/receive	Confirm receipt
POST /outcomes	Record final outcome
GET /impact/me	Personal impact
GET /admin/metrics	Admin KPIs
GET /admin/organizations	Verification list
POST /admin/rules	Update taxonomy/rules
POST /admin/disputes/{id}/resolve	Resolve dispute

21. Build folder structure

```
resavo/  
  apps/  
    mobile_flutter/  
      lib/  
        core/  
        features/auth/  
        features/seller/  
        features/buyer/  
        features/matches/  
        features/transfers/  
        features/impact/  
        services/  
    admin_web/  
      src/  
        app/  
        features/network/  
        features/impact/  
        features/verification/  
        features/disputes/  
  backend/  
    app/  
    api/  
    domain/  
    services/  
    models/  
    repositories/  
    ml/  
    jobs/  
    security/  
    tests/  
  infra/  
    docker/  
    migrations/  
    ci/  
  docs/  
    product_spec.md  
    api_contract.md  
    safety_rules.md
```

22. Testing and acceptance criteria

- User can register and select one or more roles without duplicate accounts.
- Seller can create an offer in under one minute for a supported item.
- Buyer can create a need with quantity, deadline, purpose and fulfillment preference.
- Match engine explains why each candidate is recommended or rejected.
- Engine can explicitly return "do not transfer" when a closer alternative is better.
- Pickup and receipt have separate verification events.
- Only verified outcomes enter impact calculations.
- Admin sees network growth and verified impact separately.
- Exact household location is not exposed publicly before acceptance.
- Food-specific safety rules prevent unsupported safety claims.
- Offline capture queues events and syncs after connectivity returns.
- All critical actions have audit logs.
- ML models can be replaced without rewriting the mobile UI.
- All recommendation outputs have a reason/evidence block.
- Core user flow works on a low-end Android device with slow connectivity.

22.1 Non-functional targets

Area	Target
Performance	Common screens should feel responsive on low-end Android devices; heavy operations run asynchronously.
Reliability	No silent loss of offer/need/transfer events; retry and idempotency for API writes.
Accessibility	Large touch targets, readable text, simple language, voice-ready architecture.
Security	RBAC, secure token handling, encrypted transport, audit logs.
Scalability	Stateless API services, queue-backed jobs, indexed geospatial queries.

23. Master AI build prompt

Copy the following section as the master instruction to an AI coding assistant. It is intentionally explicit so the AI understands the exact product instead of inventing unrelated features.

You are building RESAVO, a people-first public-benefit marketplace and resource-preservation network.

PRODUCT PURPOSE

RESAVO helps people and organizations offer useful resources, helps real buyers/receivers find what they genuinely need, and uses intelligence to determine whether a transfer is actually practical. The system is NOT primarily an ecommerce marketplace and must never force a transaction simply because two users match.

PRIMARY PORTALS - EXACTLY THREE

1. SELLER / PROVIDER mobile portal: create offers, set sale/free/donation/exchange terms, manage offers, accept/reject matches, manage handover and see verified impact.
2. BUYER / RECEIVER mobile + responsive web portal: create needs, compare supply options, accept offers, manage receipt and confirm outcome. Buyer types include family, business, bakery, restaurant, NGO, school, hostel, institution and community organization.
3. ADMIN / IMPACT web console: verify organizations, moderate listings, manage rules/taxonomy, resolve disputes, monitor network growth and display verified impact. Do not expose raw household histories.

PHASE 1 ITEMS - DO NOT ADD ELECTRICITY

Food: vegetables, fruits, milk/dairy, bakery/packaged food, staples/ingredients, eligible prepared surplus food.

Clothing: shirts, trousers, dresses, uniforms, jackets.

Warmth/bedding: blankets, sweaters, bedsheets.

Footwear: shoes and sandals.

Education: books, notebooks, stationery, school bags.

Household reusable: utensils, containers, storage items, non-electrical furniture.

Business/institutional surplus: non-hazardous reusable stock and supplies.

Future modules may include electricity, water and other resources, but they are OUT OF SCOPE for Phase 1.

OFFER MODES

SELL, LOW-PRICE SELL, FREE, DONATE, EXCHANGE, COMMUNITY TRANSFER.

CORE RULE

Do not optimize for maximum number of transactions. Optimize for practical preservation of useful value.

MATCHING FLOW

OFFER -> UNDERSTAND -> FIND NEED -> CHECK LOCAL ALTERNATIVES -> CHECK TIME -> CHECK FULFILLMENT -> MATCH / ALTERNATIVE / POOL / WAIT / NO TRANSFER -> VERIFY -> IMPACT -> LEARN.

IMPORTANT EXAMPLE

Family has 3 kg milk 1 km from bakery. Bakery can buy from a milk shop 300 m away. RESAVO must recommend the shop and must NOT send a delivery person to the family just because the family offered milk. The algorithm is allowed and expected to say DO NOT TRANSFER.

CLOTHING EXAMPLE

Family has 20 usable clothes. NGO can buy the lot at a low price for redistribution. The system must treat NGO as a purpose-specific receiver, consider seller preference, quantity, condition, distance, fulfillment and social purpose, and present the recommendation transparently.

TECH STACK

Mobile: Flutter + Dart.

Web admin: Next.js + React + TypeScript.

Backend: FastAPI + Python.

Database: PostgreSQL + PostGIS.

Cache/jobs: Redis.

Storage: S3-compatible object storage.

Routing: OpenStreetMap-compatible routing engine.

Notifications: FCM/APNs with SMS fallback where needed.

Deployment: Docker + cloud containers/VMs.

KEY SERVICES

AuthService, OfferService, NeedService, ItemService, MatchService, FeasibilityService, TransferService, VerificationService, ImpactService, OrganizationVerificationService, DisputeService, NotificationService, AdminService.

MATCH SCORING INPUTS

Need fit, quantity fit, time window, distance, route burden, existing local alternatives, offer mode, buyer purpose, pickup capability, reliability, safety/handling rules, confidence.

SYSTEM MUST EXPLAIN EVERY RECOMMENDATION

Show a reason such as: 'Recommended because buyer needs 3 kg by 5 PM, seller is 0.8 km away, no closer verified supply is available, and buyer can pick up.'

TRANSFER RULES

Prefer buyer pickup or seller drop when already convenient. Prefer existing routes over new dedicated trips. Allow pooled pickup when multiple offers can be consolidated. Use verified transfer partners only when required. Never create a transfer if the practical burden outweighs the benefit.

VERIFICATION

Pickup confirmation and receipt confirmation are separate. Impact is counted only after verified outcome. Partial completion must be supported.

ADMIN METRICS

Active users, active sellers, active buyers, organizations, offers, needs, successful matches, failed transfers, verified value preserved, quantity preserved, community needs fulfilled, unnecessary transfers avoided, system health.

UI STYLE

Public utility, not flashy AI. Warm off-white background, deep navy primary, green impact, blue information, amber caution, red alert. No heavy gradients. One primary action per screen.

DATA MODELS

users, profiles, offers, needs, items, item_evidence, matches, transfers, verifications, impact_events, organizations, disputes, audit_logs, model_runs.

OFFLINE-FIRST

Seller/buyer can draft offers and needs without internet. Queue events locally and sync safely later. Never duplicate an offer because of a retry.

SAFETY

Do not claim food safety purely from image. Keep safety rules configurable and conservative. Do not support hazardous items. Do not add electricity in Phase 1.

DELIVERABLES

1. Flutter mobile app with Seller and Buyer role modes.
2. Next.js Admin / Impact console.
3. FastAPI backend with OpenAPI docs.
4. PostgreSQL schema + migrations.
5. Redis job queue.
6. Seed data for milk, vegetables, clothes, books, NGO and bakery examples.
7. Unit tests, API tests and end-to-end tests.
8. Docker Compose for local development.
9. README with exact setup commands.
10. Screens that match the RESAVO UI specification.
11. No placeholder logic for the core match/feasibility algorithm; implement a deterministic baseline first.
12. Keep ML behind a service interface so it can be swapped later.

24. Future expansion

The long-term system can grow without changing its core decision loop. Future resource modules are added as resource-specific sensing, rules and models, while the offer/need/match/transfer/verify/impact infrastructure remains shared.

Future stage	Expansion
Stage 2	More food types, more NGOs, community kitchens, schools and local business categories.
Stage 3	Water and meter-based utilities when reliable integrations are available.
Stage 4	Household appliance and non-hazardous material reuse.
Stage 5	City-level resource intelligence and public-impact planning.
Stage 6	Cross-city and national resource networks.
Stage 7	Advanced resource-specific AI and global configuration.

24.1 Electricity - explicitly future

Electricity is intentionally not part of Phase 1. Future versions may accept electricity bills, meter photos, smart-meter feeds and optional smart plugs, but that module should be designed separately and should never instruct users to perform dangerous electrical work.

25. Final product definition

Operating principle: RESAVO is a public-benefit system. It is not designed around advertising, paid priority, transaction commissions or selling user data. Product decisions should be judged by whether they increase practical resource preservation and access for people and communities.

RESAVO is the new project created from the SIH 26234 food-resource problem and the REGAIN closed-loop resource-loss architecture. The first release is a three-portal public-benefit network: Seller/Provider, Buyer/Receiver and Admin/Impact. It supports practical, everyday categories such as vegetables, milk, bakery and packaged food, clothing, blankets, footwear, books, school supplies, household reusable goods and non-hazardous business surplus.

Its most important technical principle is that a match is not enough. RESAVO must compare alternatives and real fulfillment constraints. A family with 3 kg milk does not get a needless delivery trip to a bakery when a shop 300 m away can serve the bakery better. Conversely, when an NGO genuinely needs 20 clothes and can receive them efficiently, the system can connect that purpose-specific need to a family offer.

The product is designed to be helpful first and intelligent second in appearance, while being technically sophisticated underneath: explainable matching, geospatial feasibility, role-based portals, offline-safe synchronization, verification, impact measurement and continuous learning from outcomes.

North-star: Help useful resources reach their best practical next use - and count only what the system can verify.

Reference foundations and scope notes

Foundation 1: SIH 2026 Problem Statement 26234, Ministry of Food Processing Industries - AI-powered smart food waste reduction and sustainable redistribution ecosystem for institutional kitchens and food processing units (user-provided brief).

Foundation 2: REGAIN OS Mega Project Blueprint (user-provided PDF). Its key mechanisms used here include the closed product loop, risk scoring, intervention hierarchy, outcome verification, recovery-value measurement, resource passport and phased expansion. Citation tokens are intentionally not embedded in the artifact; the content is based on the supplied project material.

Naming note: RESAVO is a working coined name selected for this specification because it suggests resource + save/reuse. Before any public release, perform a formal trademark, company-name, app-store and domain clearance.